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(54) **DRIVE UNIT**

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B60K 17/12 (2006.01)
H02K 11/33 (2016.01)

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CPC **H02K 11/0094** (2013.01); **B60K 17/12** (2013.01); **H02K 11/33** (2016.01)

(58) **Field of Classification Search**
CPC H02K 11/0094; H02K 11/33; H02K 7/116;
H02K 11/30; H02K 11/02; B60K 17/12;
B60K 1/00; B60K 2001/001; F16H
37/082; H02M 1/44; H02P 27/06
See application file for complete search history.

(56) **References Cited**

U.S. PATENT DOCUMENTS

2018/0363755	A1	12/2018	Wakui	
2019/0084403	A1 *	3/2019	Nishikawa	B60K 17/105
2021/0129654	A1 *	5/2021	Krause	B60K 6/405
2021/0320566	A1 *	10/2021	Yamamoto	H02K 5/06

FOREIGN PATENT DOCUMENTS

JP	2017099096	A	6/2017
JP	2019001399	A	1/2019
JP	2020-078971	A	5/2020
JP	2021168562	A	10/2021

* cited by examiner

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(57) **ABSTRACT**

A drive unit includes: a transaxle case that houses a motor, a speed reducer, and a differential mechanism; a power control unit that controls the motor; and a noise filter electrically connected to the motor via the power control unit, the power control unit being integrated with the transaxle case, the motor, the speed reducer, and the differential mechanism being disposed on a same axis inside the transaxle case. Further, the power control unit is separate from the noise filter, and the noise filter is integrated with the transaxle case in a state in which at least a part of the noise filter is disposed in space between the power control unit and the motor.

3 Claims, 8 Drawing Sheets

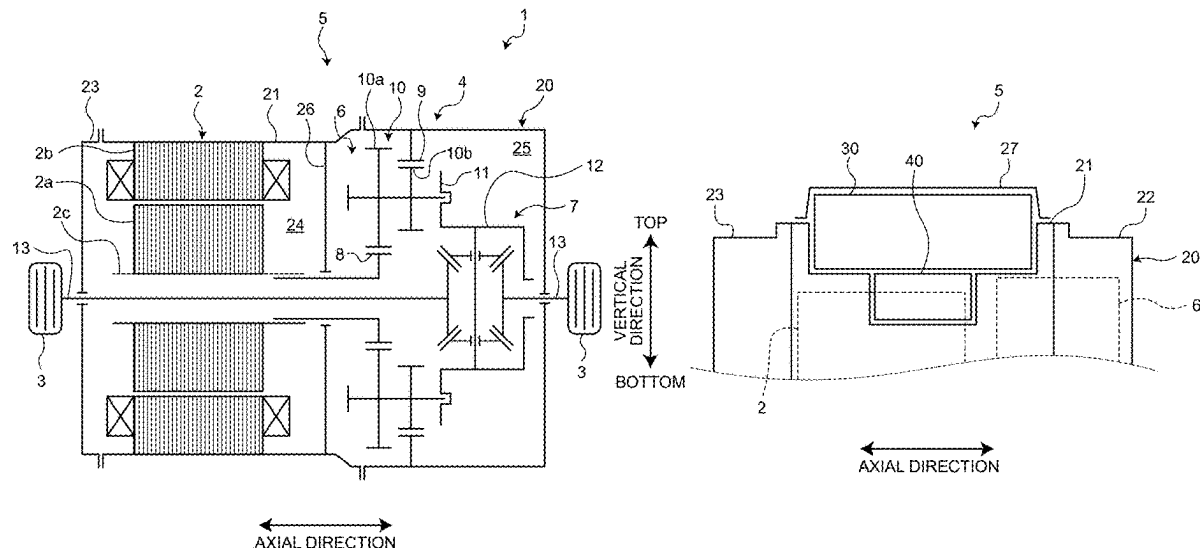


FIG.1

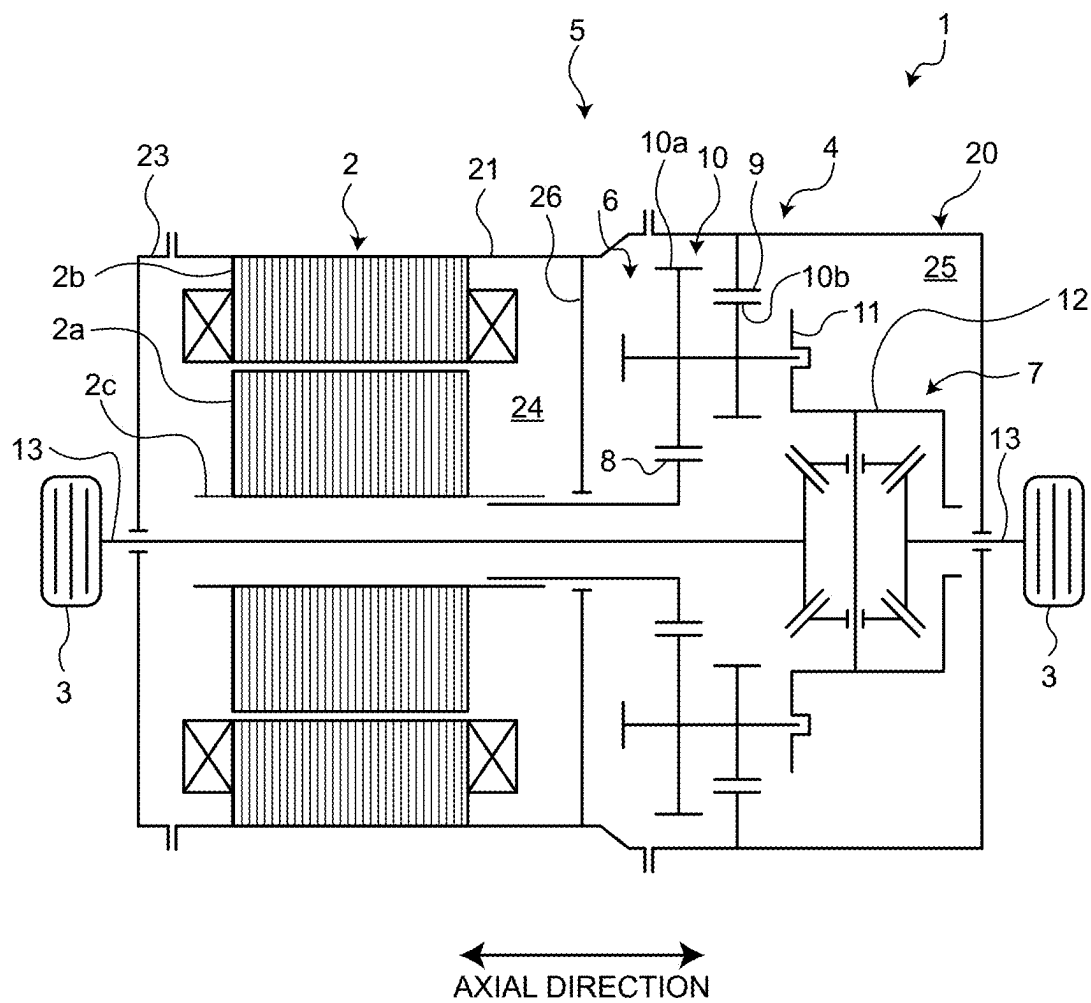


FIG. 2

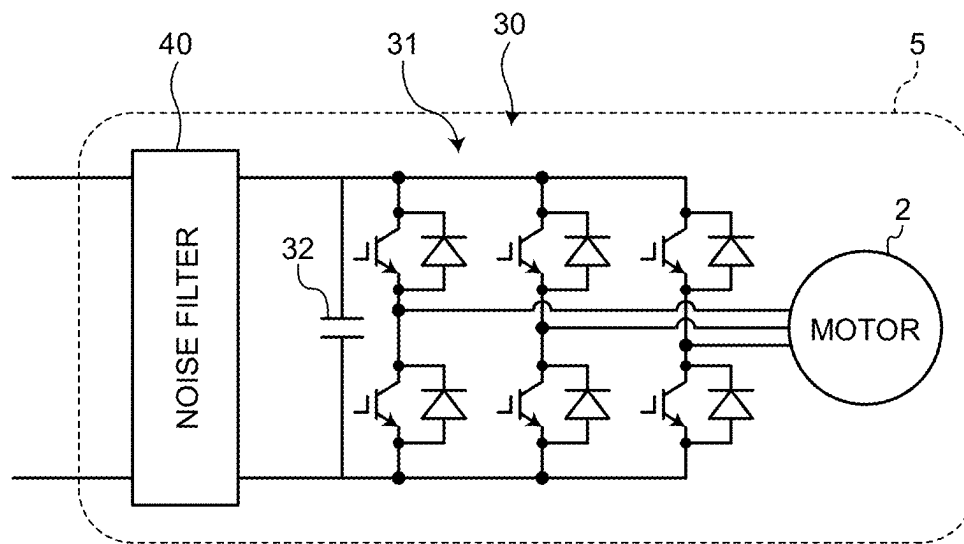


FIG.3

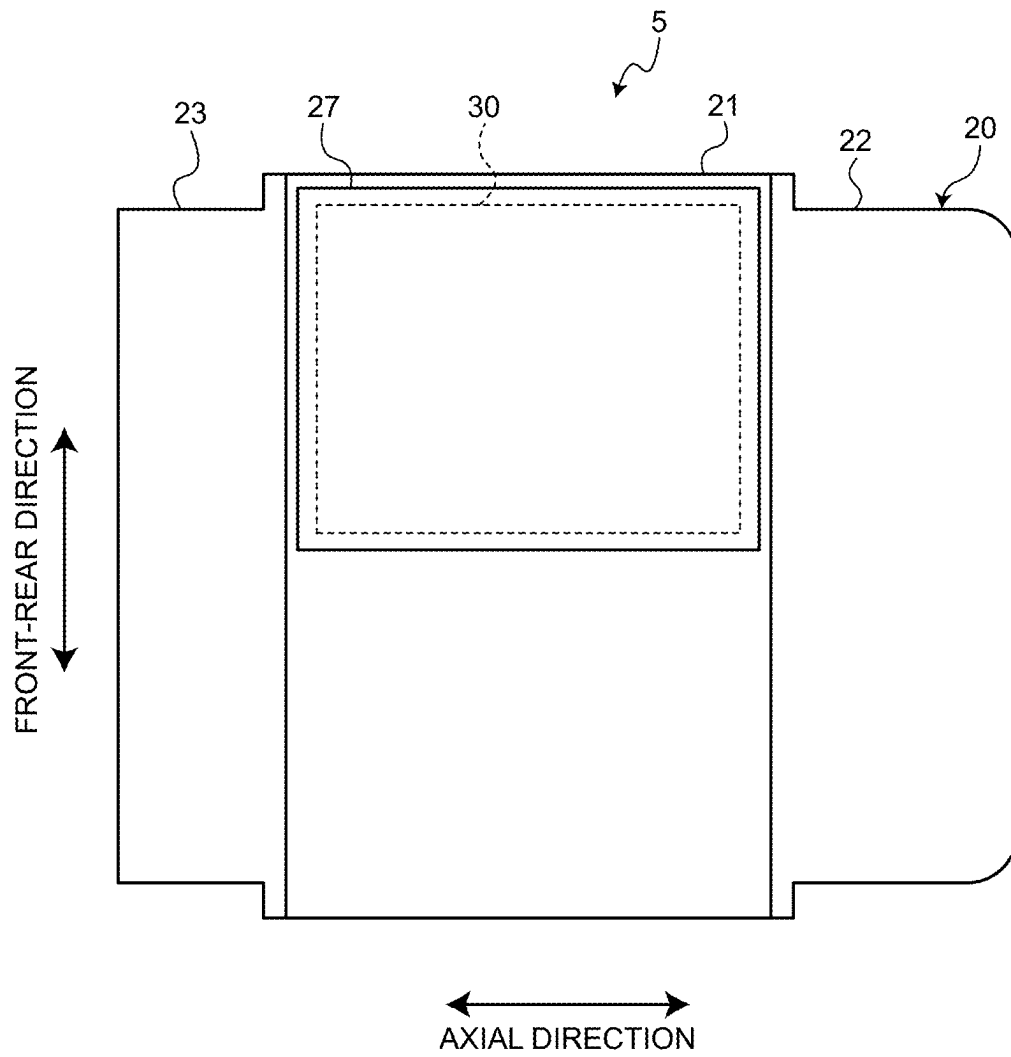


FIG.4

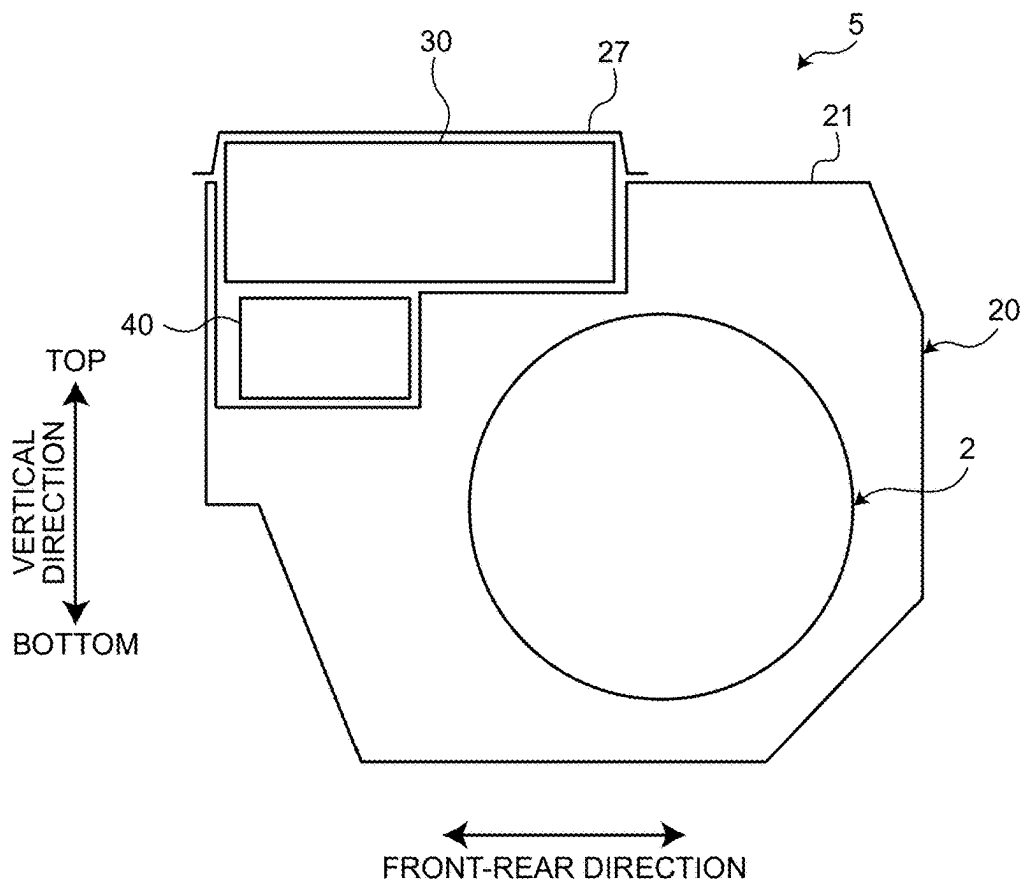


FIG.5

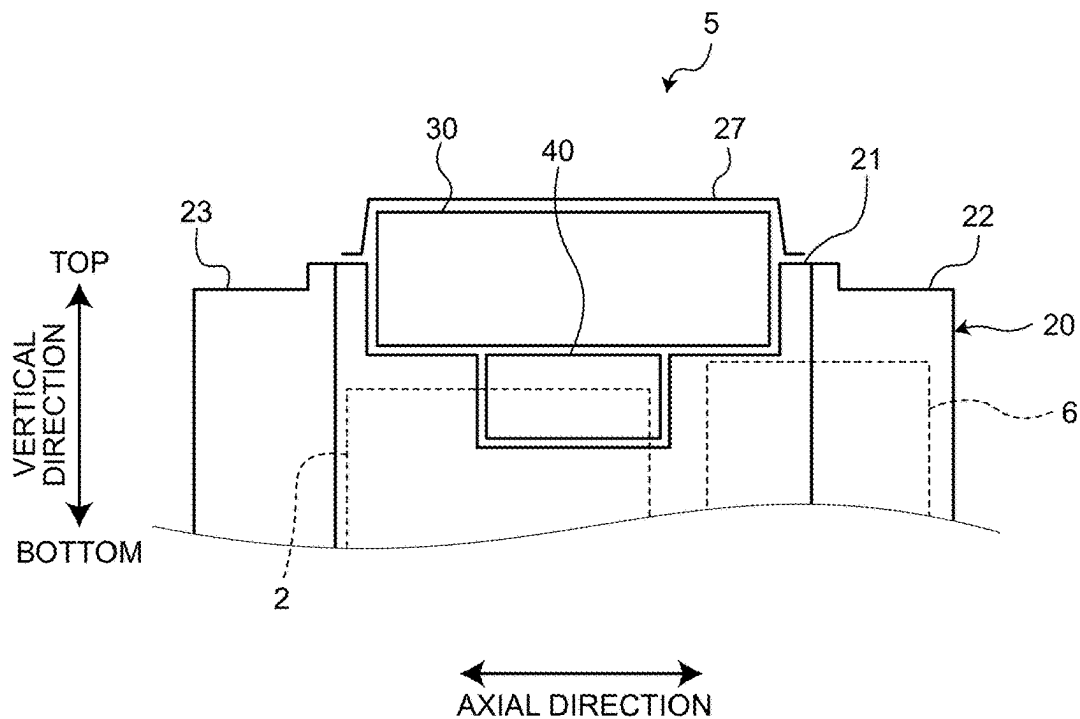


FIG. 6

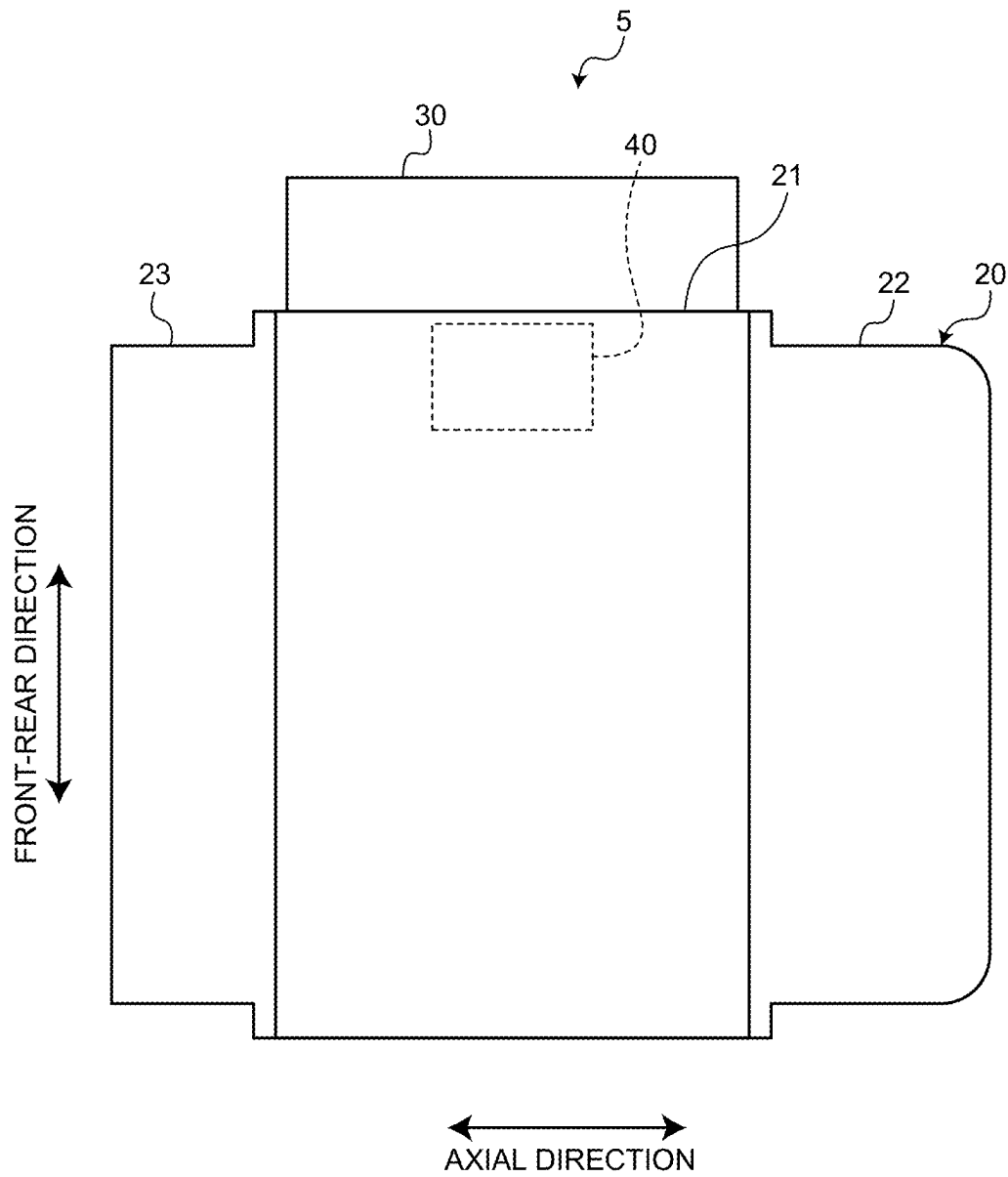


FIG. 7

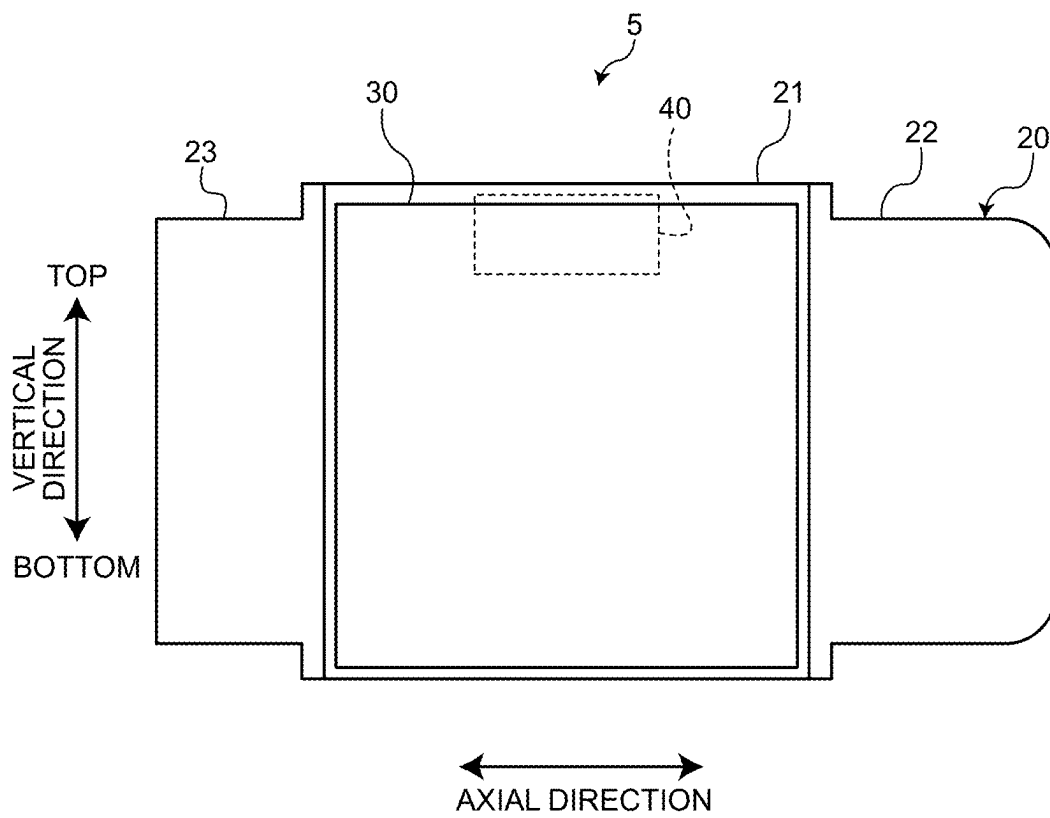


FIG.8

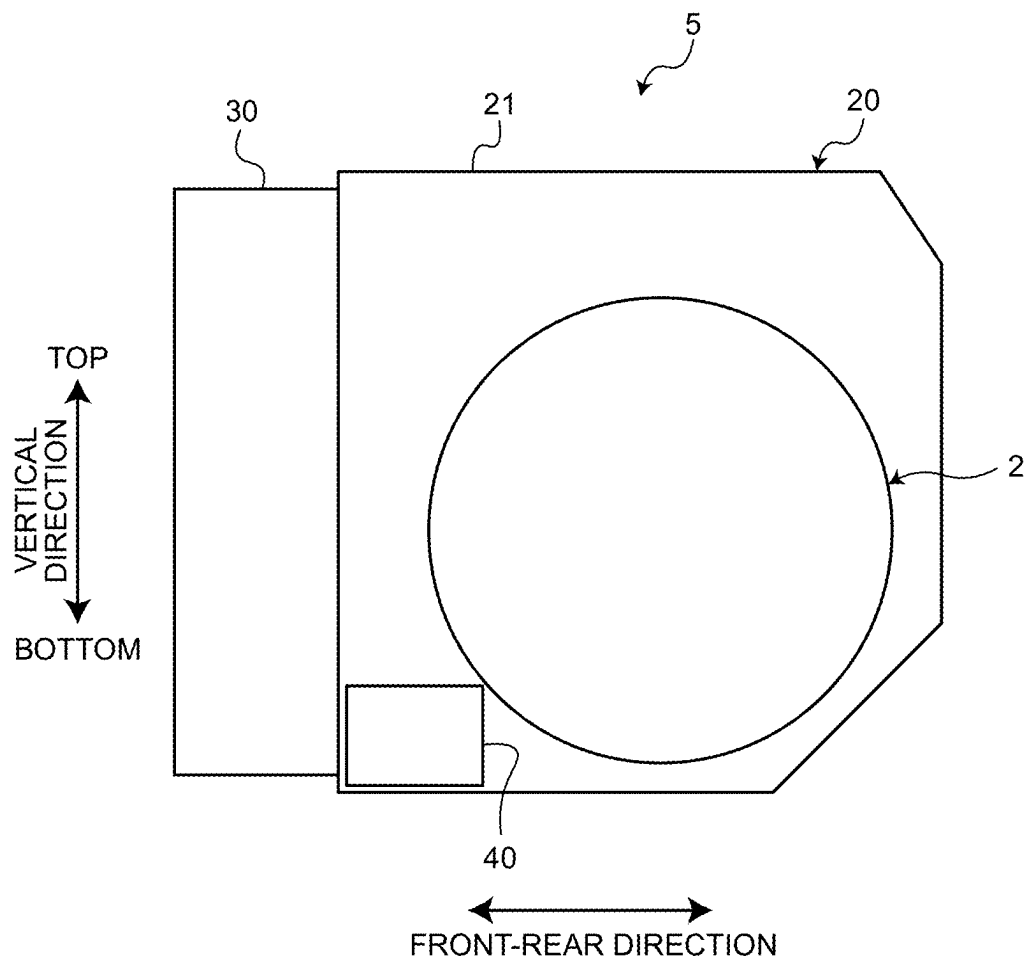
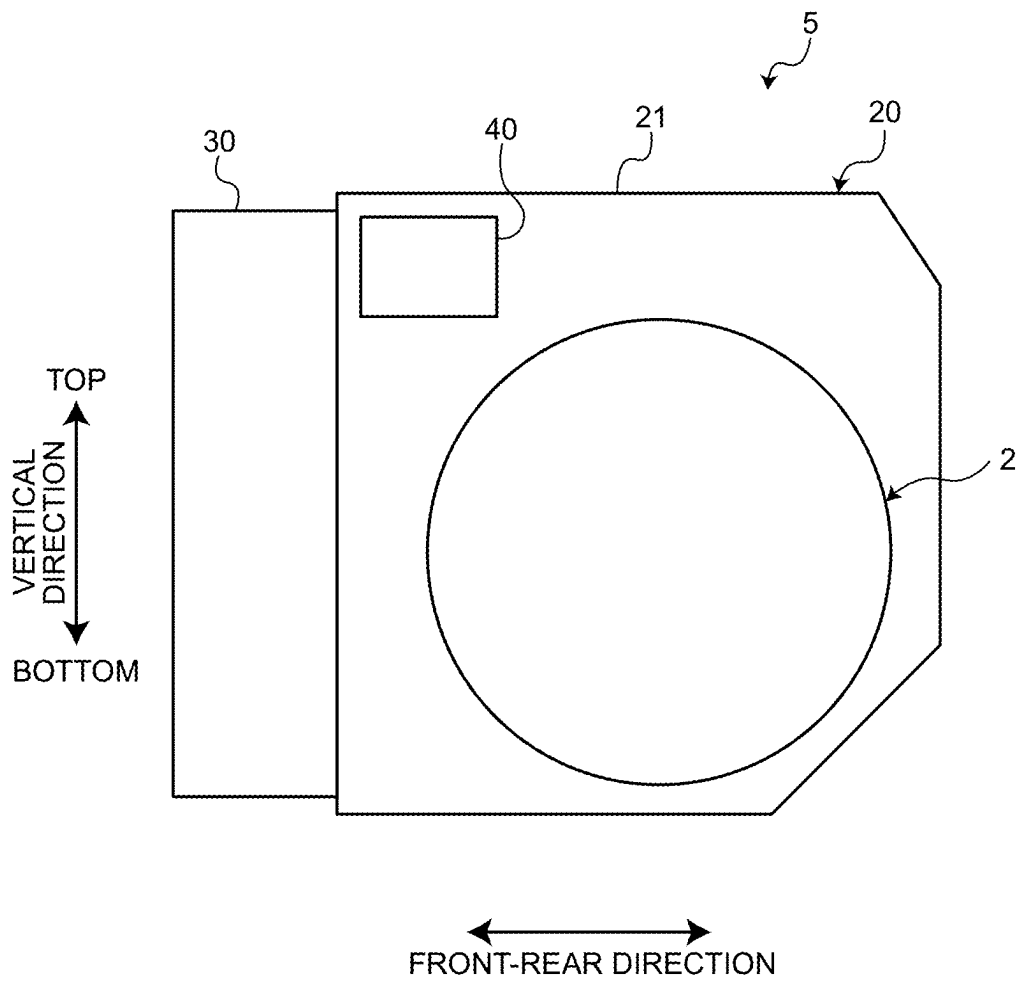


FIG.9



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DRIVE UNIT

CROSS-REFERENCE TO RELATED APPLICATION(S)

The present application claims priority to and incorporates by reference the entire contents of Japanese Patent Application No. 2022-138456 filed in Japan on Aug. 31, 2022.

BACKGROUND

The present disclosure relates to a drive unit.

Japanese Laid-open Patent Publication NO. 2020-078971 discloses a structure of a drive unit of an electric vehicle, in which a power control unit for controlling a motor is integrated with a case for housing the motor, a speed reducer, and a differential mechanism (electromechanical integrated structure). In the configuration described in Japanese Laid-open Patent Publication NO. 2020-078971, a transaxle having a triaxial structure, in which the motor, the speed reducer, and the differential mechanism are disposed on different axes, is formed.

SUMMARY

There is a need for providing a drive unit capable of efficiently disposing a power control unit and a noise filter in an electromechanical integrated structure.

According to an embodiment, a drive unit includes: a transaxle case that houses a motor, a speed reducer, and a differential mechanism; a power control unit that controls the motor; and a noise filter electrically connected to the motor via the power control unit, the power control unit being integrated with the transaxle case, the motor, the speed reducer, and the differential mechanism being disposed on a same axis inside the transaxle case. Further, the power control unit is separate from the noise filter, and the noise filter is integrated with the transaxle case in a state in which at least a part of the noise filter is disposed in space between the power control unit and the motor.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1 schematically illustrates a vehicle according to an embodiment;

FIG. 2 illustrates an electric circuit included in a drive unit;

FIG. 3 schematically illustrates the drive unit formed in an electromechanical integrated structure;

FIG. 4 illustrates space for installing a PCU and a noise filter;

FIG. 5 illustrates the arrangement of the PCU and the noise filter;

FIG. 6 schematically illustrates a drive unit according to a variation;

FIG. 7 illustrates the arrangement of a PCU and a noise filter according to the variation;

FIG. 8 illustrates the arrangement of the noise filter according to the variation; and

FIG. 9 illustrates the arrangement of a noise filter according to another variation.

DETAILED DESCRIPTION

In the related art, in addition to a transaxle having a triaxial structure, there is known a transaxle having a uni-

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axial structure in which a motor, a speed reducer, and a differential mechanism are disposed on the same axis. The transaxle having a uniaxial structure can be downsized as compared with the transaxle having a triaxial structure.

Therefore, an electromechanical integrated structure is conceivable for a drive unit including a transaxle having a uniaxial structure.

In the related art, since a power control unit needs to satisfy a standard related to electromagnetic compatibility (EMC), an electric circuit in which a noise filter and the power control unit are electrically connected is formed for addressing EMC noise. Therefore, a structure of the drive unit including the transaxle having a uniaxial structure is considered in which a noise filter is provided and integrated with a case for housing the transaxle having a uniaxial structure.

The size of the case for housing the transaxle having a uniaxial structure is, however, smaller than the size of a case for housing the transaxle having a triaxial structure. As a result, if the electromechanical integrated structure is formed, it becomes difficult to secure space for installing the power control unit in a case, and there arises a need for further securing space for installing the noise filter.

A drive unit according to an embodiment of the present disclosure will be specifically described below with reference to the drawings. Note that the present disclosure is not limited to the embodiment described below.

FIG. 1 schematically illustrates a vehicle according to the embodiment. A vehicle **1** is an electric vehicle mounted with a motor **2** serving as a power source. The vehicle **1** includes the motor **2** for traveling and a power transmission device **4** that transmits power of the motor **2** to wheels **3**. A drive unit **5** mounted in the vehicle **1** includes the motor **2**, the power transmission device **4**, and a case **20**.

The motor **2** is a motor/generator that functions as an electric motor and a generator. The motor **2** is driven by electric power supplied from a battery. The motor **2** can generate electric power by receiving torque input from the wheels **3**. The motor **2** includes a rotor **2a**, a stator **2b**, and a rotor shaft **2c**.

The rotor **2a** rotates integrally with the rotor shaft **2c**. A coil is wound around a stator core of the stator **2b**. The stator **2b** is fixed to the case **20** in a state of being disposed on the outer peripheral side of the rotor **2a**. The rotor shaft **2c** is a rotation shaft of the motor **2**. The rotor shaft **2c** is disposed so as to extend along the vehicle width direction of the vehicle **1**, and is rotatably supported by the case **20** via a bearing. An axial direction of the motor **2** is the same as the vehicle width direction of the vehicle **1**.

The power transmission device **4** forms a power transmission path between the motor **2** and the wheels **3**. The power transmission device **4** includes a speed reducer **6** and a differential mechanism **7**. The power transmission device **4** includes a transaxle having a uniaxial structure in which the motor **2**, the speed reducer **6**, and the differential mechanism **7** are disposed on the same axis.

When transmitting the power input from the motor **2** to the differential mechanism **7**, the speed reducer **6** reduces rotation input from the motor **2**, and outputs the rotation to the differential mechanism **7**. The speed reducer **6** includes a planetary gear mechanism. The speed reducer **6** includes a sun gear **8**, a ring gear **9**, and a carrier **11**. The ring gear **9** is disposed concentrically with the sun gear **8**. The carrier **11** rotatably and revolvably holds a pinion gear **10** meshed with the sun gear **8** and the ring gear **9**. The pinion gear **10** is a stepped pinion gear having a large diameter portion **10a** and a small diameter portion **10b**.

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The rotor **2a** is coupled to the sun gear **8**. Since a rotation shaft of the sun gear **8** is spline-fitted to the rotor shaft **2c** of the motor **2**, the sun gear **8** rotates integrally with the rotor shaft **2c**. In the pinion gear **10**, the large diameter portion **10a** meshes with the sun gear **8**, and the small diameter portion **10b** meshes with the ring gear **9**. The ring gear **9** is fixed to the case **20**. The wheels **3** are coupled to the carrier **11** via the differential mechanism **7**. The carrier **11** rotates integrally with a differential case **12** of the differential mechanism **7**. The sun gear **8** is an input element. The ring gear **9** is a reaction element. The carrier **11** is an output element.

The differential mechanism **7** is a differential device that allows differential rotation of a pair of right and left axles **13**. The differential mechanism **7** includes the differential case **12**, which is a rotating member. The differential case **12** rotates integrally with the carrier **11** of the speed reducer **6**. The differential mechanism **7** transmits the power transmitted from the motor **2** via the speed reducer **6** to the wheels **3** while allowing the difference between rotation speeds of the pair of right and left axles **13**.

The case **20** is a transaxle case that houses the motor **2**, the speed reducer **6**, and the differential mechanism **7**. In the case **20**, the motor **2**, the speed reducer **6**, and the differential mechanism **7** are disposed on the same axis. The case **20** houses a transaxle having a uniaxial structure. The case **20** includes a first case member **21**, a second case member **22**, and a cover member **23**.

The first case member **21** functions as a motor case, and has a shape in which both axial sides are opened. The second case member **22** functions as a gear case, and is attached to one axial end of the first case member **21**. The cover member **23** functions as a part of the motor case, and is attached to the other axial end of the first case member **21**. The first case member **21** and the second case member **22** are integrated by bolt fastening. The first case member **21** and the cover member **23** are integrated by bolt fastening.

In the case **20**, the cover member **23**, the first case member **21**, and the second case member **22** are disposed in this order from the other side to one side in the axial direction of the motor **2**, and a motor chamber **24** and a gear chamber **25** are formed. The motor chamber **24** is space for housing the motor **2**. The gear chamber **25** is space for housing the speed reducer **6** and the differential mechanism **7**.

The motor chamber **24** includes the cover member **23** and the first case member **21**. The gear chamber **25** includes the first case member **21** and the second case member **22**. The first case member **21** internally includes a partition wall **26**. The partition wall **26** separates the motor chamber **24** from the gear chamber **25**.

FIG. 2 illustrates an electric circuit included in the drive unit. The drive unit **5** includes a power control unit (hereinafter referred to as PCU) **30** and a noise filter **40**.

The PCU **30** controls the motor **2**. The PCU **30** includes an inverter **31** that drives the motor **2**. The inverter **31** is a power conversion device that converts DC power supplied from the battery into AC power, and supplies the AC power to the motor **2**. The inverter **31** includes a plurality of switching elements, and performs a switching operation under the control of an electronic control device.

The noise filter **40** is electrically connected to the PCU **30**. The noise filter **40** is provided in an electric circuit between the battery and the PCU **30**, and reduces noise when electric power from the battery is supplied to the inverter **31**. The noise filter **40** reduces, for example, EMC noise. In the electric circuit, the noise filter **40** is provided on a positive electrode line and a negative electrode line. In the electric circuit, a smoothing capacitor **32** is provided between the

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noise filter **40** and the inverter **31**. The smoothing capacitor **32** smooths the electric power output from the noise filter **40**, and supplies the power to the inverter **31**.

Here, the arrangement of the PCU **30** and the noise filter **40** in the drive unit **5** having an electromechanical integrated structure will be described with reference to FIGS. 3 to 5.

The drive unit **5** has a structure in which the PCU **30** is integrated with the case **20** (electromechanical integrated structure). In the drive unit **5** having the electromechanical integrated structure, the noise filter **40** is integrated with the case **20** in addition to the PCU **30**.

Furthermore, since the drive unit **5** includes a transaxle having a uniaxial structure, the horizontal size of the case **20** is smaller than the size of a case that houses a transaxle having a triaxial structure in a plane orthogonal to the axial direction of the motor **2** as illustrated in FIG. 4. This can reduce the size, but makes it difficult to secure space for installing the PCU **30** and the noise filter **40**. Therefore, in the drive unit **5**, the PCU **30** and the noise filter **40** are disposed with high space efficiency.

When the PCU **30** and the noise filter **40** are attached to the case that houses a transaxle having a triaxial structure, the differential mechanism is disposed on an axis different from the axis of the motor and at a position different from the axial position of the motor, so that, as a comparison, installation space for the PCU **30** and the noise filter **40** is easily secured near the motor by utilizing space on an axis of the differential mechanism in the plane orthogonal to the axial direction of the motor. In contrast, the case **20** that houses a transaxle having a uniaxial structure is smaller in size than that having a triaxial structure. Thus, if the PCU **30** and the noise filter **40** are disposed at positions similar to those in the triaxial structure, the drive unit **5** has an irregular entire shape. Therefore, the drive unit **5** is configured to increase the space efficiency while inhibiting the imbalance of the entire shape.

Specifically, the PCU **30** and the noise filter **40** are attached to the first case member **21** of the case **20**. The PCU **30** is separate from the noise filter **40**. Since the noise filter **40** needs to be connected to the positive electrode line and the negative electrode line, the noise filter **40** needs to be disposed at a position close to the PCU **30**. The noise filter **40** is disposed on the back surface side of the PCU **30**, and is disposed in space between the motor **2** and the PCU **30**. The noise filter **40** is disposed in the space sandwiched between the motor **2** and the PCU **30**.

The PCU **30** is attached to an upper portion of the first case member **21** in a state of being housed inside the PCU case. That is, the PCU case is attached to the first case member **21**. A cover member **27** is attached to an upper portion of the case **20**. Space formed by the case **20** and the cover member **27** is used for installing the PCU **30** and the noise filter **40**. The noise filter **40** is integrated with the first case member **21** in a state of being disposed in space sandwiched between the back surface of the PCU case and the first case member **21**.

In the case **20**, an oil pipe is disposed in space above the motor chamber **24**. The oil pipe is disposed immediately above the stator **2b**, and supplies oil for cooling the motor **2** to an upper portion of the stator **2b**. That is, since the space for housing the PCU **30** is restricted by the oil pipe, the space is formed at a position horizontally shifted from immediately above the motor **2**. Moreover, in the case **20**, a catch tank is disposed in space above the gear chamber **25**. The catch tank stores oil in the gear chamber **25**, and catches oil

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scraped up by rotations of the speed reducer 6 and the differential mechanism 7. The oil stored in the catch tank is supplied to the oil pipe.

Therefore, the noise filter 40 is disposed in the space between the motor 2 and the PCU 30 in a radial direction of the motor 2. In the axial direction of the motor 2, the PCU 30 is disposed at a position where the PCU 30 overlaps a part of the motor 2 and a part of the speed reducer 6, and the noise filter 40 is disposed at a position where the noise filter 40 overlaps the motor 2.

Furthermore, the noise filter 40 is disposed in space sandwiched between a connection portion and the speed reducer 6. The connection portion electrically connects the motor 2 and the PCU 30 with each other. The connection portion is a member that electrically connects the motor 2 and the inverter 31 with each other, and includes a three-phase connection portion including bus bars of respective phases. For example, the three-phase connection portion includes a connection base on which a three-phase bus bar is installed. The three-phase bus bar electrically connects a U-phase coil, a V-phase coil, and a W-phase coil of the motor 2 with the inverter 31.

The aspect ratio of the PCU 30 cannot be freely changed. The PCU 30 has a lower degree of freedom than the noise filter 40. Therefore, a location where the PCU 30 can be installed is easily restricted to a place having a large plane area. Then, since the first case member 21 forming the motor chamber 24 can have the largest area in the case 20, the PCU 30 is attached to the first case member 21.

The noise filter 40 has a higher degree of freedom in shape than the PCU 30. Therefore, the connection portion between the motor 2 and the PCU 30 and the gear chamber 25 are located on both sides of a housing portion of the noise filter 40. Space for installing the noise filter 40 can be secured even in a place where the installation space is easily restricted by shape, that is, a place where the axial length cannot be easily secured in the case 20.

As described above, in the drive unit 5 having the electromechanical integrated structure, the space efficiency can be enhanced by installing the noise filter 40 in the space between the motor 2 and the PCU 30. As a result, in the drive unit 5, the transaxle having a uniaxial structure can save space while space for mounting the noise filter 40 in the electromechanical integrated structure can be secured. Furthermore, since the noise filter 40 is separate from the PCU 30, the noise filter 40 can be mounted even if the noise filter 40 is enlarged.

Furthermore, the capacity of the catch tank can be increased in the gear chamber 25 inside the second case member 22 by attaching the PCU 30 and the noise filter 40 to the first case member 21.

Note that the PCU 30 is not required to be entirely attached to the first case member 21. Similarly, the noise filter 40 is not required to be entirely attached to the first case member 21. Only at least parts of the PCU 30 and the noise filter 40 are required to be attached to the first case member 21. The remaining part may be attached to the second case member 22 and the cover member 23.

Furthermore, the cover member 27 is not necessarily provided. That is, the present disclosure is not limited to a structure in which the PCU 30 and the noise filter 40 are housed in closed space formed by the cover member 27.

Furthermore, the noise filter 40 is not required to be entirely disposed in the space between the PCU 30 and the motor 2. That is, the noise filter 40 is only required to be

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attached to the case 20 in a state in which at least a part of the noise filter 40 is disposed in the space between the PCU 30 and the motor 2.

Furthermore, the noise filter 40 is not required to be entirely disposed in space between the connection portion electrically connecting the motor 2 and the PCU 30 with each other and the speed reducer 6. That is, only at least a part of the noise filter 40 is required to be disposed in the space between the connection portion electrically connecting the motor 2 and the PCU 30 with each other and the speed reducer 6.

Furthermore, installation places of the PCU 30 and the noise filter 40 are not limited to an upper portion of the case 20. The PCU 30 and the noise filter 40 can be disposed at a front portion, a rear portion, and a lower portion of the case 20. FIGS. 6 to 9 illustrate one example of variations.

As illustrated in FIGS. 6 to 8, the PCU 30 is attached to the front portion of the case 20 in the drive unit 5 according to a variation. The noise filter 40 is disposed in the space between the PCU 30 and the motor 2 in the front-rear direction of the vehicle 1, and is disposed below the rotation center position of the motor 2 in the vertical direction of the vehicle 1.

As illustrated in FIG. 9, the noise filter 40 may be disposed above the rotation center position of the motor 2 in the vertical direction of the vehicle 1 in the drive unit 5 according to another variation.

In the present disclosure, a power control unit and a noise filter can be efficiently disposed in an electromechanical integrated structure.

According to an embodiment, a power control unit and a noise filter can be efficiently disposed in an electromechanical integrated structure.

According to an embodiment, the noise filter can be disposed with high space efficiency.

According to an embodiment, the noise filter can be disposed with high space efficiency.

Although the disclosure has been described with respect to specific embodiments for a complete and clear disclosure, the appended claims are not to be thus limited but are to be construed as embodying all modifications and alternative constructions that may occur to one skilled in the art that fairly fall within the basic teaching herein set forth.

What is claimed is:

1. A drive unit comprising:

a transaxle case that houses a motor, a speed reducer, and a differential mechanism;

a power control unit that controls the motor; and

a noise filter electrically connected to the motor via the power control unit,

the power control unit being integrated with the transaxle case, the motor, the speed reducer, and the differential mechanism being disposed on a same axis inside the transaxle case,

wherein the power control unit is separate from the noise filter, and

the noise filter is integrated with the transaxle case in a state in which at least a part of the noise filter is disposed in space between the power control unit and the motor.

2. The drive unit according to claim 1, further comprising a connection portion that electrically connects the motor and the power control unit with each other,

wherein at least a part of the noise filter is disposed in space between the connection portion and the speed reducer.

3. The drive unit according to claim 1,
wherein the transaxle case includes:
a motor case that houses the motor; and
a gear case that houses the speed reducer and the differ-
ential mechanism, 5
wherein a PCU case that houses the power control unit is
attached to the motor case, and
the noise filter is attached to the motor case in a state of
being disposed in space between a back surface of the
PCU case and the motor case. 10

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